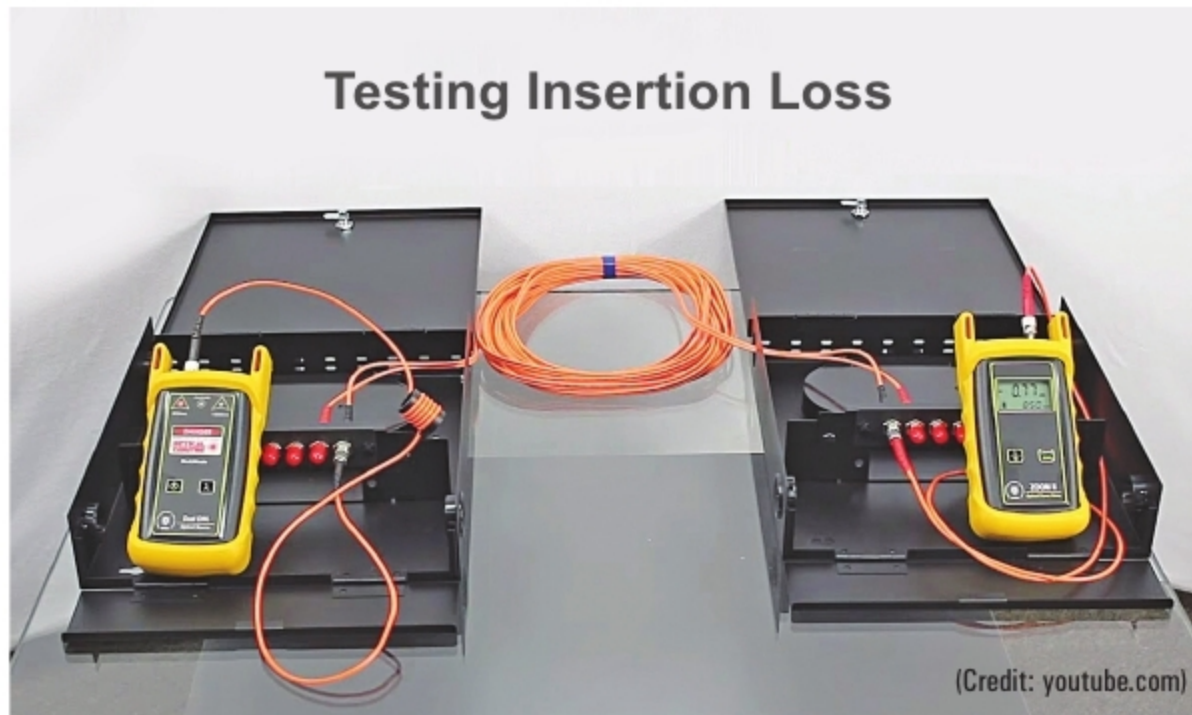


Optical-Fibre Cable Testing Practices And Challenges

Testing evaluates the performance of fibre-optic components, cable plants and systems. For every fibre-optic cable plant, testing for continuity and polarity, end-to-end insertion loss and then troubleshooting any problems is necessary.



An optical-fibre cable, also known as fibre-optic cable, is like an electrical cable but contains one or more optical fibres that are used to carry light (signals). An optical-fibre cable consists of a core and a cladding layer. Signal transmission takes place in this type of cable due to total internal reflection, which happens due to the difference in the refractive index between the fibre material and air or any other surrounding material.

In September 2012, NTT Japan demonstrated a single fibre cable that was able to transfer one petabit per second over 50 kilometres. A single, modern fibre cable contains up to 1000 fibres, with tetrabytes per second of potential bandwidth.

Today, optical communication is preferred over wireless transmission because of its large bandwidth and low propagation attenuation. Also, it costs less.

Advantages of an optical-fibre cable are:

- **Speed:** A fibre-optic cable network operates at a high speed, in gigabytes.
- **Bandwidth:** It provides large computing capacity.
- **Distance:** Signals can be transmitted without needing to be refreshed or strengthened.
- **Resistance:** It offers greater resistance to electromagnetic noise from radios, motors or other nearby cables.
- **Maintenance:** It requires extremely low maintenance.
- It is less bulky, lighter and more flexible than conventional cables.
- It can carry large amounts of data. Its applications include:
 - Internet
 - Cable TV
 - Telephone
 - Computer network
 - Surgery and dentistry
 - Lighting and decorations
 - Mechanical inspection

- Military and space applications
- Automobile industry

Accurate testing of cables ensures that no good cables are rejected, or bad ones are accepted. Industries spend a lot of time and effort to develop better standards that ensure accurate testing. But these standards are generally written for manufacturers and not customers.

One of the measurements required to confirm the quality of installation is optical loss or insertion loss of each fibre in the cable. Loss of measurement is made on end-to-end basis in a permanently-installed cable plant. For this, a test source and optical power meter—sometimes called optical loss test set—and reference set cables are used. Sometimes, an optical time-domain reflectometer is also used.

The right tools for accurate optical-fibre cable testing

A lot of time and cost can be saved if the contractor and installers know the proper measurements that need to be done, how to make these measurements correctly, which tools to use and keep these tools in good condition—tools must be calibrated from time to time.

All errors that occur in an optical-fibre cable can be broadly classified into random errors and systematic errors.

Fibre attenuation measurement is essential. Attenuation is the loss of optical power as a result of absorption, scattering, bending and other loss mechanisms as light travels through fibre. Cut-back method is often used for measuring total attenuation of the optical-fibre cable. Modal bandwidth testing in a multi-mode optical-fibre cable is also essential.